

Yogesh Gandhi

PhD in Aerospace Engineering

Computational Mechanics · Topology Optimization · Composites

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Computational scientist with a PhD in aerospace engineering and seven years of research on the geometry-first end of computational mechanics — from CAD-based topology optimization of fiber-reinforced composites to nonlinear FE infrastructure for industrial simulation. First-author in CMAME; production work shipped on parts for Ferrari and the winning Sasol 2024 solar vehicle. Currently building the simulation engine behind automated engineering design at Braid Technologies, Tokyo.

Technical skills

- **Languages:** Python · MATLAB · Fortran · Bash
- **Numerics & FE:** FEM/FEA (linear, nonlinear, implicit, explicit) · contact · buckling · dynamic explicit · IGA / NURBS · corotational shells · multiscale modeling
- **Optimization:** topology optimization (SIMP, geometry projection, CAD-based) · gradient-based methods · manufacturing constraints · multi-load problems
- **Solvers & tools:** LS-DYNA · Kratos Multiphysics · MFEM · Abaqus · ANSYS · SolidWorks · GMSH · Markforged · 9tlabs
- **Systems:** Linux · Git · LaTeX · scientific Python stack (NumPy, SciPy, JAX)
- **Spoken:** English (C1) · Italian (B2) · Japanese (learning)

Experience

Senior Researcher — Braid Technologies K.K., Tokyo, Japan

Aug 2025 – Present

- Develop the mathematical models and FE building blocks that power Braid's automated design engine, serving Tier-1 automotive and manufacturing clients.
- Built and integrated nonlinear-static, contact, and constraint modules; debugged convergence in 3D implicit buckling with simultaneous geometric, material, and contact nonlinearities.
- Implemented new element formulations (thin shells, corotational kinematics) and orchestrate multi-solver strategies across LS-DYNA, Kratos Multiphysics, and MFEM.
- Maintain mesh-processing pipelines spanning TRI3/TRI6, hex, and IGA-ready geometries.

Postdoctoral Researcher — University of Bologna, Forlì, Italy

Jan – Jun 2024

- Translated PhD-stage methods into production CFRP design workflows in the ASCENT and OptiFiber projects.
- Delivered stress-based topology optimization for 3D-printed wing-mechanism parts of the Ferrari 296GTB.
- Co-led design and additive manufacturing of structural parts for the Emilia V solar vehicle — winner of the Sasol 2024 Cruiser Challenge.

Research Associate — University of Parma, Parma, Italy

Apr 2018 – Feb 2020

- Implemented thermo-mechanical UMAT subroutines (Abaqus) for shape memory alloys; derived fiber-layup optimization for bistable laminates.
- Built VCCT and cohesive-element delamination models calibrated against experimental data.

Earlier: Visiting Graduate Researcher, Coventry University, UK (2017) — path-planning (A*, D*) and non-linear control (Particle Swarm Optimization) for autonomous rovers. Research collaborations with TU Delft,

Politecnico di Torino, and Università di Padova (2021–2025) — CAD-based topology optimization, Carrera Unified Formulation for laminates, and 9tlabs additive process modeling.

Education

PhD, Aerospace Engineering — University of Bologna, Forlì, Italy 2020 – 2024

Geometry projection method for additively manufactured variable-stiffness continuous fiber-reinforced polymer structures.

MSc, Aerospace Engineering — University of Bologna, Forlì, Italy 2015 – 2017

BEng, Mechanical Engineering — PES Institute of Technology, Bengaluru, India 2010 – 2014

Selected projects

- **Geometry projection for variable-stiffness composites.** Unified topology-optimization framework for additively manufactured composite laminates with continuous fibers; designs layout, orientation, and stiffness in one formulation. *CMAME*, 2025.
- **Ferrari 296GTB wing mechanism (OptiFiber).** Stress-based design framework for 3D-printed composite parts; topology-optimization methodology and experimental validation.
- **Emilia V solar vehicle.** Co-led structural design and Markforged 3D printing of components; first place at the Sasol 2024 Cruiser Challenge, South Africa.
- **TO with Carrera Unified Formulation (with POLITO).** Extended TO via CUF for accurate through-thickness modeling of multilayered laminates.

Selected publication

126 citations · h-index 5 · i10-index 4 (*Google Scholar*, 2026)

Y. Gandhi, A. Aragón, J. Norato, G. Minak. *A geometry projection method for the topology optimization of additively manufactured variable-stiffness composite laminates.* *Computer Methods in Applied Mechanics and Engineering*, 2025. doi:10.1016/j.cma.2024.117663

Full publication list at gandhiyudheer.github.io/#publications and ORCID.

Teaching, mentoring & service

- PhD co-supervisor, University of Bologna (2024 – present) — design framework for 3D-printed parts using Emilia V experimental data. MSc co-supervisor (2024) — stress-based TO for AM structures.
- Lecturer: MSc Mechanical Eng, University of Bologna (2023, 2024) — topology-optimization theory & MATLAB; MSc Automotive Eng, Dallara Academy (2019) — composite failure & delamination in Abaqus.
- Peer reviewer for *Composite Structures* and *Additive Manufacturing*.